

Srivatsan Manikandan

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EDUCATION

Rutgers University-New Brunswick

Bachelor of Science in Computer Science, Minor in Business Administration

GPA: 3.65/4.0

- **Relevant Coursework:** Introduction to Computer Science, Data Structures, Computer Architecture, Intro to Linear Algebra, Systems Programming, Discrete Structures, Data Management for Data Science, Physics 1, Physics 2, Design and Algorithms, Software Methodology, Principles of Information and Data Management

EXPERIENCE

ING Group | Software Engineer Intern

New York, NY | **May 2025 - Present**

- Developed a Python-based analytics tool to process daily cybersecurity reports retrieved via an internal ING API connected to the Amsterdam database.
- Automated the extraction and visualization of device-level vulnerability data from large Excel datasets, eliminating the need for manual review of thousands of records.
- Delivered clear, actionable dashboards that enable the cybersecurity team to quickly identify and respond to at-risk devices.

Robert Wood Johnson Hospital | Software Engineer Intern

New Brunswick, NJ | **June 2024 - Present**

- Engineered a Python-based application to enhance ureter visualization during surgeries, reducing operation time by an average of 10 minutes and improving postoperative outcomes.
- Leveraged advanced physics algorithms to compute precise robotic blade movements relative to ureters, minimizing accidental injuries and lowering surgical complication rates by 7%.
- Designed an intuitive front-end interface to display color-coded visualizations, highlighting scalpel proximity to ureters

Rutgers Office of Information and Technology | IT Consultant

New Brunswick, NJ | **April 2024 - Present**

- Implemented robust encryption standards and multi-factor authentication protocols to secure sensitive student information, mitigating potential data breaches and enhancing cybersecurity compliance.
- Delivered prompt and effective technical support to students, faculty, and staff, maintaining optimal system functionality

Steam Works Studio | Developer

Princeton, NJ | **June 2022 - August 2022**

- Developed cost-efficient alternatives to commercial educational kits by integrating Arduino with Lego Technic components.
- Created a Python script of 100+ lines to enable seamless communication between Arduino devices and Lego Technic parts.
- Modeled 3D building designs in AutoCAD for use in educational workshops focused on electrical engineering concepts

PROJECTS

Robert Wood Johnson Hospital (BiliaryX - in development) | Python

- Designed and implemented a YOLO-based deep learning model for real-time detection of biliary ducts and veins during surgeries, enhancing precision and reducing risks of incorrect incisions.
- Conducted model training on medical imaging datasets, achieving high accuracy and robust performance in a surgical environment
- Achieved 4th place in Rutgers Health Hackathon, demonstrating innovative application of AI in healthcare under time constraints

Spell Check | C++

- Developed a spell-check program leveraging POSIX function for efficient file system operations and word parsing.
- Implemented direct word comparisons to identify and suggest corrections for misspelled words accurately
- Optimized the algorithm to handle large datasets with minimal latency, ensuring quick and reliable spell-check results

AI-Assisted 3D Model | Python, Java, Git

- Developed software to create a full 3D model from a picture using artificial intelligence
- Designed using an API that was incorporated into our Python code
- Incorporated GitHub for easy collaboration and allowed us to view edits easily.

Shell Program | C++

- Developed a custom shell for compiling C++ code, utilizing GCC, GDB, and Makefile
- Implemented features such as directory parsing, wildcard searches, and redirection

CERTIFICATIONS

AWS - Cloud Practitioner

June 2023 - Present

- Developed and optimized cloud-based applications. Implemented security measures for these cloud applications

IBM - Cybersecurity fundamentals

July 2023 - Present

- Learned about threat prevention, penetration testing, and network security

SKILLS

- **Software Tools:** Excel, Visual Studio Code, Eclipse, AWS, Git, AutoCAD
- **Programming:** Java, Kotlin, Python, HTML, C, C++, Full-Stack, React